Lecture Notes: January 27, 2026

March 23, 2026

BlackBoard: Random Walks (Jan. 27, 2026)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $(X_n)_{n\geq 1}$ be a sequence of i.i.d. random variables, and let S_0 be independent of $(X_n)_{n\geq 1}$.

$$S_n = S_0 + \sum_{j=1}^{n} X_j, \qquad n = 1, 2, \dots$$

We call $(S_n)_{n\geq 0}$ a *random walk*.

Here "independent" means S_0 is independent of the whole sequence $(X_n)_{n\geq 1}$: knowing S_0 gives no information about the increments. Formally, for any event A depending only on S_0 and any event B depending only on $(X_1, X_2, \dots)$, Here $A \in \sigma(S_0)$ (i.e., A is determined by S_0) and $B \in \sigma(X_1, X_2, \dots)$. $\mathbb{P}(A \cap B) = \mathbb{P}(A)\,\mathbb{P}(B)$. Equivalently, for any bounded/measurable functions f, g, $\mathbb{E}[f(S_0)\, g(X_1, \dots, X_n)] = \mathbb{E}[f(S_0)]\,\mathbb{E}[g(X_1, \dots, X_n)]$ for all n.

Simple Random Walk (SRW)

Special Case: Simple Random Walk

$$X_n \in \{\pm e_1, \dots, \pm e_d\}, \qquad \mathbb{P}(X_n = \pm e_k) = \frac{1}{2d}.$$

Typically we start at the origin:

$$\mathbb{P}(S_0 = 0) = 1.$$

SRW in Finite Time

Fix $N \in \mathbb{N}$. Let $\Omega_N = \{-1, 1\}^N$ with

$$X_n(\omega) = \omega_n, \qquad \mathbb{P}(A) = \frac{|A|}{2^N}.$$

Here A is any event (subset) $A \subseteq \Omega_N$. The quantity $|A|$ is the number of outcomes (sequences) in A, so $\mathbb{P}$ is the uniform measure on Ω_N.

Exercise 1

Show that $X_1, \ldots, X_N$ are i.i.d. and

$$\mathbb{P}(X_i = \pm 1) = \frac{1}{2}.$$

Proof idea / sketch (Exercise 1)

Idea. The probability measure $\mathbb{P}$ is uniform on $\Omega_N = \{-1, 1\}^N$, so probabilities are just counts divided by 2^N.

Marginals. Fix i. Exactly half of the sequences have $\omega_i = 1$ and half have $\omega_i = -1$, hence $\mathbb{P}(X_i = \pm 1) = \frac{1}{2}$.

Independence. For distinct indices $i_1, \ldots, i_k$ and prescribed signs $\varepsilon_1, \ldots, \varepsilon_k \in \{-1, 1\}$, the event $\{X_{i_1} = \varepsilon_1, \ldots, X_{i_k} = \varepsilon_k\}$ fixes k coordinates and leaves $N - k$ free, so it has 2^{N-k} outcomes and probability $2^{-k} = \prod_{j=1}^{k} \frac{1}{2}$. This shows independence (and by symmetry all X_i have the same distribution).

More detail on the independence step. The key observation is that under the uniform measure on $\Omega_N = \{-1, 1\}^N$, fixing the values of k coordinates leaves exactly $N-k$ coordinates free. Since each free coordinate can be chosen in 2 ways, there are 2^{N-k} sequences consistent

with the constraints. Thus

$$\mathbb{P}(X_{i_1} = \varepsilon_1, \ldots, X_{i_k} = \varepsilon_k) = \frac{2^{N-k}}{2^N} = 2^{-k} = \prod_{j=1}^{k} \mathbb{P}(X_{i_j} = \varepsilon_j),$$

which is the definition of independence. In words: *count the number of outcomes satisfying the constraints and compare with the product of the marginal counts.*

Exercise 2

Show that SRW has stationary independent increments.

Proof idea / sketch (Exercise 2)

Key definitions.

- The (simple) random walk is $S_0 = 0$ and $S_n = \sum_{j=1}^{n} X_j$.
- An *increment* over an interval $[a, b]$ is $S_b - S_a$.
- *Independent increments* means that for $0 \leq k_0 < k_1 < \cdots < k_m \leq N$, the random variables $S_{k_1} - S_{k_0}, \ldots, S_{k_m} - S_{k_{m-1}}$ are independent.
- *Stationary increments* means the law of an increment depends only on its length: for $0 \leq k < m \leq N$, $S_m - S_k \overset{d}{=} S_{m-k}$.

Step 1: write increments as sums of disjoint blocks. Using $S_n = \sum_{j=1}^{n} X_j$, for any $0 \leq a < b \leq N$ we have the telescoping identity

$$S_b - S_a = \sum_{j=a+1}^{b} X_j.$$

In particular, if $0 \leq k_0 < k_1 < \cdots < k_m \leq N$, then for each $r = 1, \ldots, m$,

$$S_{k_r} - S_{k_{r-1}} = \sum_{j=k_{r-1}+1}^{k_r} X_j.$$

Notice that the index-sets $\{k_{r-1} + 1, \ldots, k_r\}$ are *pairwise disjoint* across different r.

Step 2: independent increments. A standard fact: if $\{X_j\}$ are independent, then any two functions of disjoint subcollections are independent. Here each increment $S_{k_r} - S_{k_{r-1}}$ depends only on the variables $X_{k_{r-1}+1}, \ldots, X_{k_r}$, and those blocks do not overlap, hence the increments are independent.

Step 3: stationarity of increments. Fix $0 \le k < m \le N$. Then

$$S_m - S_k = \sum_{j=k+1}^{m} X_j.$$

This is a sum of exactly $(m-k)$ i.i.d. $\{\pm 1\}$–valued random variables. Because the X_j's are *identically distributed*, the vector $(X_{k+1}, \ldots, X_m)$ has the same joint distribution as $(X_1, \ldots, X_{m-k})$. Therefore

$$S_m - S_k = \sum_{j=k+1}^{m} X_j \stackrel{d}{=} \sum_{j=1}^{m-k} X_j = S_{m-k},$$

which is the stationary-increments property.

Technique summary. Reduce each increment to a *block sum* of the underlying i.i.d. steps X_j. Then: *disjoint blocks* $\Rightarrow$ *independence*, and *equal-length blocks of i.i.d. variables* $\Rightarrow$ *same distribution.*

Reminder: what "independent" means. Random variables $Y_1, \ldots, Y_m$ are independent if for all Borel sets $A_1, \ldots, A_m \subseteq \mathbb{R}$,

$$\mathbb{P}(Y_1 \in A_1, \ldots, Y_m \in A_m) = \prod_{i=1}^{m} \mathbb{P}(Y_i \in A_i).$$

Equivalently (often convenient for proofs), for all bounded measurable functions $f_1, \ldots, f_m$,

$$\mathbb{E}\left[\prod_{i=1}^{m} f_i(Y_i)\right] = \prod_{i=1}^{m} \mathbb{E}[f_i(Y_i)].$$

Why "bounded measurable"? We require $f_1, \ldots, f_m$ to be measurable so that $f_i(Y_i)$ is

a well-defined random variable. We assume they are bounded to avoid integrability issues: boundedness guarantees $\prod_{i=1}^{m} f_i(Y_i)$ is integrable, so the expectation $\mathbb{E}[\prod_i f_i(Y_i)]$ is always finite and well-defined. (One can replace "bounded" by integrable conditions, but "bounded measurable" is the clean standard statement.)

How Step 2 uses this (independent increments). From Step 1 we wrote each increment as a *block sum*

$$S_{k_r} - S_{k_{r-1}} = \sum_{j=k_{r-1}+1}^{k_r} X_j.$$

So each increment is a function of the block of variables

$$(X_{k_{r-1}+1}, \ldots, X_{k_r}).$$

Different increments correspond to *disjoint* blocks of indices, and since $X_1, \ldots, X_N$ are independent, the σ-fields

$$\sigma(X_{k_0+1}, \ldots, X_{k_1}),\ \sigma(X_{k_1+1}, \ldots, X_{k_2}),\ \ldots$$

are independent. In other words: *disjoint blocks + independent coordinates (i.e., the coordinate random variables $X_1, \ldots, X_N$ are independent) $\Rightarrow$ the σ-algebras generated by each block are independent.* Therefore any functions of these blocks (in particular, the sums defining the increments) are independent.

Concretely, if you pick bounded measurable test functions $\varphi_1, \ldots, \varphi_m$ and let $Y_r := S_{k_r} - S_{k_{r-1}}$, then

$$\mathbb{E}\left[\prod_{r=1}^{m} \varphi_r(Y_r)\right] = \prod_{r=1}^{m} \mathbb{E}[\varphi_r(Y_r)],$$

which is exactly the independence criterion above.

Moments and Distribution

$$\mathbb{E}(S_n) = 0, \qquad \mathrm{Var}(S_n) = n.$$

Why $\mathrm{Var}(S_n) = n$ (for SRW with $X_j \in \{\pm 1\}$ and $\mathbb{P}(X_j = 1) = \mathbb{P}(X_j = -1) = \frac{1}{2}$). Recall $S_n = \sum_{j=1}^{n} X_j$.

Step 1: compute $\mathbb{E}[X_j]$ and $\mathrm{Var}(X_j)$. We have

$$\mathbb{E}[X_j] = 1 \cdot \tfrac{1}{2} + (-1) \cdot \tfrac{1}{2} = 0, \qquad \mathbb{E}[X_j^2] = 1, \qquad \mathrm{Var}(X_j) = \mathbb{E}[X_j^2] - (\mathbb{E}[X_j])^2 = 1.$$

Step 2: use independence to add variances. Since $X_1, \ldots, X_n$ are independent,

$$\mathrm{Var}(S_n) = \mathrm{Var}\left(\sum_{j=1}^{n} X_j\right) = \sum_{j=1}^{n} \mathrm{Var}(X_j) \quad \text{(cross-covariances are 0).}$$

Therefore

$$\mathrm{Var}(S_n) = \sum_{j=1}^{n} 1 = n.$$

Remark. More generally, if X_j are i.i.d. with $\mathrm{Var}(X_j) = \sigma^2$, then $\mathrm{Var}(S_n) = n\sigma^2$.

$$\mathbb{P}(S_n = x) = \binom{n}{\frac{n+x}{2}} 2^{-n}, \qquad x \equiv n \pmod 2 \quad \text{(this is the parity condition)}$$

$$\mathbb{P}(S_n = x) = \mathbb{P}(S_n = -x)$$

Stopping Times

Let $\mathcal{A}_n = \sigma(X_1, \ldots, X_n)$.

Definition (Stopping Time)

A random variable $T : \Omega \to \{0, 1, \ldots, N\} \cup \{\infty\}$ is a stopping time if

$$\{T = n\} \in \mathcal{A}_n.$$

Intuition. A stopping time is a random time at which you can decide to stop by looking only at the information revealed *up to that time* (no peeking into the future). Equivalently,

once you have seen $X_1, \ldots, X_n$, you can determine whether $T = n$ happened (i.e., whether the random time $T(\omega)$ takes the value n).

$$\tau_a = \inf\{n \geq 0 : S_n = a\} \quad \text{is a stopping time.}$$

Why τ_a is a stopping time. Recall $\mathcal{A}_n = \sigma(X_1, \ldots, X_n)$ and S_k is measurable with respect to $\mathcal{A}_k$ (i.e., the value of S_k can be determined using only the information available up to time k. Recall $S_k = X_1 + \cdots X_k$ so it's a function of $\{X_1, ..., X_k\}$ so once I know first k steps, I know S_k). For each $n \geq 0$ we can write

$$\{\tau_a = n\} = \{S_n = a\} \cap \bigcap_{k=0}^{n-1} \{S_k \neq a\}.$$

Each event $\{S_n = a\}$ belongs to $\mathcal{A}_n$, and for $k \leq n-1$ we have $\{S_k \neq a\} \in \mathcal{A}_k \subseteq \mathcal{A}_n$. Hence the intersection is in $\mathcal{A}_n$, so $\{\tau_a = n\} \in \mathcal{A}_n$ for all n, which is exactly the definition of a stopping time.

Optional Stopping

"If you stop a martingale at a *reasonable* random time, its expected value does not change."

Optional stopping is a special case of professor's theorem below, obtained by choosing a particular predictable strategy.

Theorem

For any predictable strategy (V_k),

$$\mathbb{E}\left(\sum_{k=1}^{N} V_k X_k\right) = 0.$$

Remark 1. $\mathbb{E}\left(\sum_{k=1}^{N} V_k X_k\right)$ is called a martingale transform. (X_k) is a martingale difference sequence (e.g., the random walk increments).

Remark 2 (Interpretation of predictable strategies and fair games). A *predictable strategy*

$(V_k)_{k=1}^N$ is a sequence of random variables such that

$$V_k \in L^1(\mathcal{A}_{k-1}), \tag{1}$$

i.e. the choice of V_k is based only on the information available strictly before time k. In the gambling interpretation, V_k represents the stake placed at time $k-1$ on the outcome of the kth step X_k.

The gain at time k is $V_k X_k$, and the total gain up to time N is

$$G_N := \sum_{k=1}^{N} V_k X_k.$$

The theorem asserts that

$$\mathbb{E}(G_N) = 0$$

for *any* predictable strategy (V_k).

This expresses the fact that the simple random walk is a *fair game*. Indeed, since X_k is independent of $\mathcal{A}_{k-1}$ and satisfies $\mathbb{E}(X_k) = 0$, we have

$$\mathbb{E}(X_k \mid \mathcal{A}_{k-1}) = 0.$$

Consequently,

$$\mathbb{E}(V_k X_k \mid \mathcal{A}_{k-1}) = V_k \, \mathbb{E}(X_k \mid \mathcal{A}_{k-1}) = 0.$$

(V_k comes out of $\mathbb{E}$ because 1). Taking expectations and summing over k yields $\mathbb{E}(G_N) = 0$.

The predictability condition is essential: if V_k were allowed to depend on X_k, one could choose $V_k = \operatorname{sign}(X_k)$ and obtain a sure gain. Thus the theorem formalizes the principle that no strategy based solely on past information can extract positive expected profit from a fair random walk.

Remark 3 (Meaning of $L^1(\mathcal{A}_{k-1})$). This condition has two components. First, V_k is $\mathcal{A}_{k-1}$–measurable, meaning that the stake chosen at time k depends only on the information available strictly before time k. Second, V_k is integrable $\mathbb{E}(|V_k|) < \infty$, which ensures that

the gain $V_k X_k$ has finite expectation, since $|X_k| = 1$, and that the expected total gain $\mathbb{E}\left(\sum_{k=1}^{N} V_k X_k\right)$ is well defined.

Corollary

If T is a bounded stopping time, then

$$\mathbb{E}(S_T) = 0, \qquad \mathrm{Var}(S_T) = \mathbb{E}(T).$$

Remark 4 (Explanation of the variance identity). Recall $S_n = \sum_{k=1}^{n} X_k$, where (X_k) are i.i.d. with $\mathbb{E}(X_k) = 0$ and $\mathbb{E}(X_k^2) = 1$. Since $\mathbb{E}(S_T) = 0$, we have $\mathrm{Var}(S_T) = \mathbb{E}(S_T^2)$.

Write

$$S_T = \sum_{k=1}^{T} X_k = \sum_{k=1}^{\infty} X_k \, \mathbf{1}_{\{k \le T\}},$$

where the sum is finite almost surely because T is bounded. Then

$$S_T^2 = \sum_{k=1}^{\infty} X_k^2 \mathbf{1}_{\{k \le T\}} + 2 \sum_{1 \le k < \ell < \infty} X_k X_\ell \, \mathbf{1}_{\{k \le T\}} \mathbf{1}_{\{\ell \le T\}}.$$

Taking expectations, the cross terms vanish. Indeed, for $k < \ell$, the random variable $I := \mathbf{1}_{\{k \le T\}} \mathbf{1}_{\{\ell \le T\}}$ is $\mathcal{A}_{\ell-1}$–measurable since $\{T \ge \ell\} = \{T \le \ell - 1\}^c \in \mathcal{A}_{\ell-1}$ since T is a stopping time and $\{T \le n\} \in \mathcal{A}_n$ for all n. Similarly, $\{T \ge k\} \in \mathcal{A}_{k-1} \subseteq \mathcal{A}_{\ell-1}$. Hence I is $\mathcal{A}_{\ell-1}$–measurable. Since $X_\ell \perp \mathcal{A}_{\ell-1}$ and $\mathbb{E}(X_\ell) = 0$, we condition on $\mathcal{A}_{\ell-1}$ and apply the tower property:

$$\mathbb{E}[X_k X_\ell I] = \mathbb{E}[\mathbb{E}(X_k X_\ell I \mid \mathcal{A}_{\ell-1})] \, .$$

Inside the conditional expectation, $X_k I$ is $\mathcal{A}_{\ell-1}$–measurable, so

$$\mathbb{E}(X_k X_\ell I \mid \mathcal{A}_{\ell-1}) = X_k I \, \mathbb{E}(X_\ell \mid \mathcal{A}_{\ell-1}) \, .$$

By independence and mean zero (see remark below for discussion),

$$\mathbb{E}(X_\ell \mid \mathcal{A}_{\ell-1}) = 0.$$

Therefore, $\mathbb{E}(X_k X_\ell I \mid \mathcal{A}_{\ell-1}) = 0$, and taking expectations again yields

$$\mathbb{E}\big(X_k X_\ell \, \mathbf{1}_{\{k \le T\}} \mathbf{1}_{\{\ell \le T\}}\big) = 0.$$

Therefore, $\mathbb{E}(S_T^2) = \sum_{k=1}^{\infty} \mathbb{E}\big(X_k^2 \mathbf{1}_{\{k \le T\}}\big)$. Since $X_k^2 = 1$ almost surely,

$$\mathbb{E}(S_T^2) = \sum_{k=1}^{\infty} \mathbb{P}(k \le T) = \mathbb{E}\left(\sum_{k=1}^{\infty} \mathbf{1}_{\{k \le T\}}\right) = \mathbb{E}(T).$$

Thus $\mathrm{Var}(S_T) = \mathbb{E}(T)$.

Remark 5 (Boundedness of T is a very strong assumption). Note that while boundedness of T implies $\mathbb{P}(T < \infty) = 1$ and $\mathbb{E}[T] < \infty$ (positive recurrence), neither "finite a.s." ($\mathbb{P}(T < \infty) = 1$ nor "finite expectation" ($\mathbb{E}[T] < \infty$) implies boundedness of T. E.g., let T be geometric, so $\mathbb{P}(T = n) = 1/2^n$ for $n \ge 1$. Then $\mathbb{E}[T] = 2 < \infty$ but for every $M, \mathbb{P}(T > M) > 0$, so T is unbounded.

Remark 6 (On the law of total expectation used above). The law of total expectation $\mathbb{E}(Y) = \mathbb{E}[\mathbb{E}(Y \mid \mathcal{G})]$ is always true, for any integrable random variable Y and any σ-algebra $\mathcal{G}$. It does not require independence. In our case, letting $Y = X_k X_\ell I$, we can always write

$$\mathbb{E}[X_k X_\ell I] = \mathbb{E}[\mathbb{E}(X_k X_\ell I \mid \mathcal{A}_{\ell-1})] \, .$$

Independence is needed for the next step, where we simplify the inner conditional expectation. We use

$$\mathbb{E}(X_\ell \mid \mathcal{A}_{\ell-1}) = 0,$$

which holds because X_ℓ is independent of $\mathcal{A}_{\ell-1}$ and $\mathbb{E}(X_\ell) = 0$.

If instead we conditioned on $\mathcal{A}_\ell$, then

$$\mathbb{E}(X_\ell \mid \mathcal{A}_\ell) = X_\ell,$$

since X_ℓ is measurable with respect to $\mathcal{A}_\ell$. In that case, the cancellation would fail.

More generally, what we are really using is the martingale property

$$\mathbb{E}(X_\ell \mid \mathcal{A}_{\ell-1}) = 0.$$

Independence is simply a sufficient condition for this property.

Remark 7 (Why we need independence and mean zero).

Case 1: Independent, mean zero (martingale difference). If X_ℓ is independent of $\mathcal{A}_{\ell-1}$ and $\mathbb{E}(X_\ell) = 0$, then

$$\mathbb{E}(X_\ell \mid \mathcal{A}_{\ell-1}) = \mathbb{E}(X_\ell) = 0.$$

Case 2: Independent, nonzero mean. If X_ℓ is independent of $\mathcal{A}_{\ell-1}$ and $\mathbb{E}(X_\ell) = \mu \neq 0$, then

$$\mathbb{E}(X_\ell \mid \mathcal{A}_{\ell-1}) = \mu.$$

In this case the conditional expectation is still deterministic.

Case 3: Dependent on the past. If X_ℓ depends on past values, then

$$\mathbb{E}(X_\ell \mid \mathcal{A}_{\ell-1})$$

is generally a nonconstant $\mathcal{A}_{\ell-1}$–measurable random variable. For example, if

$$X_\ell = X_{\ell-1} + \varepsilon_\ell,$$

where (ε_ℓ) are independent with mean zero, then

$$\mathbb{E}(X_\ell \mid \mathcal{A}_{\ell-1}) = X_{\ell-1}.$$

Remark 8 (Measurability of the indicator terms). Fix $k < \ell$. Since T is a stopping time, we have

$$\{T \leq n\} \in \mathcal{A}_n \quad \text{for all } n \geq 0.$$

Taking complements yields

$$\{T \geq n+1\} \in \mathcal{A}_n.$$

(note that this is not "peeking into the future" since it's not using X_{n+1}, but it just means that the process has not stopped by time n). In particular,

$$\{\ell \leq T\} = \{T \geq \ell\} \in \mathcal{A}_{\ell-1}.$$

Similarly, since $k \leq \ell - 1$,

$$\{k \leq T\} = \{T \geq k\} \in \mathcal{A}_{k-1} \subseteq \mathcal{A}_{\ell-1}.$$

Therefore both indicators $\mathbf{1}_{\{k \leq T\}}$ and $\mathbf{1}_{\{\ell \leq T\}}$ are $\mathcal{A}_{\ell-1}$–measurable, and so is their product.

Remark 9. Bounded T is the easiest way to guarantee the results in the above corollary, but the most common replacement is $\mathbb{E}[T] < \infty$ plus a condition (like bounded increments or uniform integrability) that prevents the martingale from "leaking mass" in the tails.

Corollary (Unbounded stopping times: sufficient conditions)

Let $S_n = \sum_{k=1}^{n} X_k$, where (X_k) are i.i.d. with $\mathbb{E}[X_1] = 0$ and $\mathbb{E}[X_1^2] = 1$, and let T be a (possibly unbounded) stopping time with respect to the natural filtration $\mathcal{A}_n := \sigma(X_1, \ldots, X_n)$. Assume in addition that T satisfies *any one* of the following standard (optional stopping / optional sampling) hypotheses:

(A) (*Uniform integrability*) The stopped martingale $(S_{n \wedge T})_{n \geq 1}$ is uniformly integrable, and $\mathbb{E}[T] < \infty$.

(B) (*Bounded increments + integrable time*) There exists $c < \infty$ such that $|X_1| \leq c$ a.s., and $\mathbb{E}[T] < \infty$.

(C) (*A common tail condition*) $\mathbb{E}[T] < \infty$ and $\lim_{n \to \infty} \mathbb{E}\big(|S_n| \, \mathbf{1}_{\{T > n\}}\big) = 0$.

Then the optional stopping conclusions hold:

$$\mathbb{E}(S_T) = 0, \qquad \mathbb{E}(S_T^2) = \mathbb{E}(T), \qquad \text{and hence} \qquad \mathrm{Var}(S_T) = \mathbb{E}(T).$$

Remark 10. (C) says that the contribution to the expectation coming from the event that

we haven't stopped yet by time n becomes negligible as $n \to \infty$. $|S_n| \, \mathbf{1}_{\{T>n\}}$ is $|S_n|$ on paths where we are still running at time n and 0 on on paths where we already stopped by time n. So as n grows, either $\{T > n\}$ becomes rare enough or $|S_n|$ on that event is controlled enough, or both, that this expected tail contribution goes to 0.

Definition 1 (Uniform integrability)**.** A family of integrable random variables $\mathcal{X} \subset L^1(\Omega, \mathcal{F}, \mathcal{P})$ is *uniformly integrable* if

$$\lim_{K\to\infty} \sup_{X\in\mathcal{X}} \mathbb{E}\big[\, |X| \, \mathbf{1}_{\{|X|>K\}} \,\big] = 0.$$

Remark 11 (Equivalent characterization)*.* A family $\mathcal{X} \subset L^1$ is uniformly integrable iff

1. $\sup_{X\in\mathcal{X}} \mathbb{E}|X| < \infty$, and
2. for every $\varepsilon > 0$ there exists $\delta > 0$ such that whenever $A \in \mathcal{F}$ with $\mathcal{P}(A) < \delta$, one has

$$\sup_{X\in\mathcal{X}} \mathbb{E}[\, |X| \, \mathbf{1}_A \,] < \varepsilon.$$

Remark 12 (Sufficient condition)*.* If there exists $p > 1$ such that

$$\sup_{X\in\mathcal{X}} \mathbb{E}\big[|X|^p\big] < \infty,$$

then $\mathcal{X}$ is uniformly integrable.

*Proof under **(A)** (uniform integrability).* Let $(\mathcal{A}_n)_{n\geq 0}$ be the natural filtration $\mathcal{A}_n := \sigma(X_1, \ldots, X_n)$. Then $(S_n)_{n\geq 0}$ is an $(\mathcal{A}_n)$–martingale with $S_0 = 0$.

Assume **(A)** and, in addition (as is standard when proving second-moment identities), that the stopped martingale

$$M_n := S_n^2 - n \qquad \text{satisfies that} \qquad (M_{n\wedge T})_{n\geq 0} \text{ is uniformly integrable.}$$

(Equivalently: optional stopping applies to both S and M at time T.)

Step 1: $\mathbb{E}[S_T] = 0$*.* By optional stopping for the uniformly integrable martingale $(S_{n\wedge T})$,

$$\mathbb{E}[S_T] = \mathbb{E}[S_0] = 0.$$

Step 2: $\mathbb{E}[S_T^2] = \mathbb{E}[T]$. Since $M_n = S_n^2 - n$ is an $(\mathcal{A}_n)$–martingale and $(M_{n\wedge T})$ is uniformly integrable, optional stopping yields

$$\mathbb{E}[M_T] = \mathbb{E}[M_0] = 0.$$

Thus $\mathbb{E}[S_T^2 - T] = 0$, i.e.

$$\mathbb{E}[S_T^2] = \mathbb{E}[T].$$

Since $\mathbb{E}[S_T] = 0$, we have $\mathrm{Var}(S_T) = \mathbb{E}[S_T^2] = \mathbb{E}[T]$. □

Proof under ***(B)*** *(bounded increments + integrable time).* Assume $|X_1| \le c$ a.s. and $\mathbb{E}[T] < \infty$.

Step 1: $\mathbb{E}[S_T] = 0$. For each $m \in \mathbb{N}$, define the bounded stopping time $T_m := T \wedge m$. By the bounded stopping time case,

$$\mathbb{E}[S_{T_m}] = 0 \quad \text{for all } m.$$

Moreover, since $|S_{T_m}| \le \sum_{k=1}^{T_m} |X_k| \le c\,T_m \le c\,T$, we have $|S_{T_m}| \le cT$ with $\mathbb{E}[cT] < \infty$. Also $S_{T_m} \to S_T$ a.s. as $m \to \infty$. By dominated convergence,

$$\mathbb{E}[S_T] = \lim_{m\to\infty} \mathbb{E}[S_{T_m}] = 0.$$

Step 2: $\mathbb{E}[S_T^2] = \mathbb{E}[T]$ *(one extra integrability assumption).* To justify passing second moments to the limit, assume additionally that $\mathbb{E}[T^2] < \infty$ (which is a common strengthening in this bounded-increments setting). Then $S_{T_m}^2 \le (cT)^2$ and $\mathbb{E}[(cT)^2] < \infty$, and again $S_{T_m}^2 \to S_T^2$ a.s. So dominated convergence gives

$$\mathbb{E}[S_T^2] = \lim_{m\to\infty} \mathbb{E}[S_{T_m}^2].$$

Using the bounded stopping time identity for each T_m,

$$\mathbb{E}[S_{T_m}^2] = \mathbb{E}[T_m].$$

Finally $T_m \uparrow T$, so by monotone convergence,

$$\lim_{m\to\infty} \mathbb{E}[T_m] = \mathbb{E}[T].$$

Combining,

$$\mathbb{E}[S_T^2] = \mathbb{E}[T].$$

Since $\mathbb{E}[S_T] = 0$, it follows that $\mathrm{Var}(S_T) = \mathbb{E}[T]$. $\square$

*Proof under **(C)** (tail condition).* Assume $\mathbb{E}[T] < \infty$ and

$$\lim_{n\to\infty} \mathbb{E}\big(|S_n|\, \mathbf{1}_{\{T>n\}}\big) = 0.$$

Step 1: $\mathbb{E}[S_T] = 0$. For each n, write

$$S_{T\wedge n} = S_T\, \mathbf{1}_{\{T\le n\}} + S_n\, \mathbf{1}_{\{T>n\}}.$$

Taking expectations and using the bounded stopping time case for $T \wedge n$,

$$0 = \mathbb{E}[S_{T\wedge n}] = \mathbb{E}\big(S_T\, \mathbf{1}_{\{T\le n\}}\big) + \mathbb{E}\big(S_n\, \mathbf{1}_{\{T>n\}}\big)\,.$$

Rearrange:

$$\mathbb{E}\big(S_T\, \mathbf{1}_{\{T\le n\}}\big) = -\mathbb{E}\big(S_n\, \mathbf{1}_{\{T>n\}}\big)\,.$$

Letting $n \to \infty$, we have $S_T\, \mathbf{1}_{\{T\le n\}} \to S_T$ a.s., and the right-hand side tends to 0 by the tail condition (using $|\mathbb{E}(S_n \mathbf{1}_{\{T>n\}})| \le \mathbb{E}(|S_n| \mathbf{1}_{\{T>n\}})$). Thus,

$$\mathbb{E}[S_T] = 0.$$

Step 2: $\mathbb{E}[S_T^2] = \mathbb{E}[T]$ *(analogous tail condition for $S_n^2 - n$).* For the second-moment identity, apply the same argument to the martingale

$$M_n := S_n^2 - n,$$

but this requires the corresponding tail condition

$$\lim_{n\to\infty} \mathbb{E}\big(|M_n|\,\mathbf{1}_{\{T>n\}}\big) = 0.$$

(Under many common hypotheses this follows from moment bounds; alternatively, you may assume it directly.) Then, exactly as above,

$$0 = \mathbb{E}[M_{T\wedge n}] = \mathbb{E}\big(M_T\,\mathbf{1}_{\{T\le n\}}\big) + \mathbb{E}\big(M_n\,\mathbf{1}_{\{T>n\}}\big)\,,$$

so letting $n \to \infty$ gives $\mathbb{E}[M_T] = 0$, i.e.

$$\mathbb{E}[S_T^2 - T] = 0 \quad\Longrightarrow\quad \mathbb{E}[S_T^2] = \mathbb{E}[T].$$

Together with $\mathbb{E}[S_T] = 0$, this yields $\mathrm{Var}(S_T) = \mathbb{E}[T]$. □

Remark 13. One standard form of the Optional Stopping Theorem says:

If $(M_n)_{n\ge 0}$ is an integrable martingale and τ is a bounded stopping time, then

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0].$$

Theorem 1 (Optional Stopping Theorem (bounded case))**.** *Let $(M_n, \mathcal{F}_n)_{n\ge 0}$ be a martingale with $\mathbb{E}|M_n| < \infty$ for all n. If τ is a bounded stopping time (i.e. $\tau \le N$ a.s. for some $N < \infty$), then M_τ is integrable and*

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0].$$

Theorem 2 (Optional Stopping Theorem (one common unbounded version))**.** *Let $(M_n, \mathcal{F}_n)_{n\ge 0}$ be a martingale and let τ be a stopping time. If the family $\{M_{\tau\wedge n} : n \ge 1\}$ is uniformly integrable and $\mathbb{E}|M_\tau| < \infty$, then*

$$\mathbb{E}[M_\tau] = \mathbb{E}[M_0].$$

Proof of OST (bounded case). Let $(M_n, \mathcal{F}_n)_{n\ge 0}$ be a martingale and let τ be a bounded stopping time, so $\tau \le N$ a.s. for some $N \in \mathcal{N}$.

Write

$$M_\tau = \sum_{k=0}^{N} M_k \, \mathbf{1}_{\{\tau=k\}}.$$

Taking expectations,

$$\mathbb{E}[M_\tau] = \sum_{k=0}^{N} \mathbb{E}\big(M_k \, \mathbf{1}_{\{\tau=k\}}\big) \, .$$

Fix $k \in \{0, 1, \ldots, N\}$. Since $\{\tau = k\} \in \mathcal{F}_k$ and $M_k = \mathbb{E}(M_N \mid \mathcal{F}_k)$ (martingale property iterated),

$$\mathbb{E}\big(M_k \, \mathbf{1}_{\{\tau=k\}}\big) = \mathbb{E}\big(\mathbb{E}(M_N \mid \mathcal{F}_k) \, \mathbf{1}_{\{\tau=k\}}\big) = \mathbb{E}\big(M_N \, \mathbf{1}_{\{\tau=k\}}\big) \, ,$$

where we used the defining property of conditional expectation for the $\mathcal{F}_k$–measurable random variable $\mathbf{1}_{\{\tau=k\}}$. Therefore,

$$\mathbb{E}[M_\tau] = \sum_{k=0}^{N} \mathbb{E}\big(M_N \, \mathbf{1}_{\{\tau=k\}}\big) = \mathbb{E}\left(M_N \sum_{k=0}^{N} \mathbf{1}_{\{\tau=k\}} \right) = \mathbb{E}[M_N].$$

Finally, since (M_n) is a martingale, $\mathbb{E}[M_N] = \mathbb{E}[M_0]$, hence $\mathbb{E}[M_\tau] = \mathbb{E}[M_0]$. $\square$

Proof of OST (uniform integrability version). Let $(M_n, \mathcal{F}_n)_{n\geq 0}$ be a martingale and let τ be a stopping time. Assume that the family $\{M_{\tau\wedge n} : n \geq 1\}$ is uniformly integrable and that M_τ is integrable.

Step 1: Apply the bounded case to $\tau \wedge n$. For each n, the stopping time $\tau \wedge n$ is bounded by n, so by the bounded OST,

$$\mathbb{E}[M_{\tau\wedge n}] = \mathbb{E}[M_0], \qquad \forall n \geq 1.$$

Step 2: Pass to the limit. Since $\tau \wedge n \uparrow \tau$, we have $M_{\tau\wedge n} \to M_\tau$ a.s. as $n \to \infty$. By Vitali's theorem: a.s. convergence + uniform integrability $\Rightarrow$ L^1 convergence. (Note that a.s. convergence alone does not guarantee convergence of expectations since mass can "escape to infinity" on rare events and UI is precisely a condition that prevents this); hence

$$\lim_{n\to\infty} \mathbb{E}[M_{\tau\wedge n}] = \mathbb{E}[M_\tau].$$

Taking limits in the identity from Step 1 gives

$$\mathbb{E}[M_\tau] = \lim_{n\to\infty} \mathbb{E}[M_{\tau\wedge n}] = \mathbb{E}[M_0].$$

□

Theorem 3 (Vitali's Convergence Theorem)**.** *Let $(X_n)_{n\geq 1}$ be a sequence of integrable random variables on $(\Omega, \mathcal{F}, \mathcal{P})$ and suppose that $X_n \to X$ almost surely. If the family $\{X_n : n \geq 1\}$ is uniformly integrable, then X is integrable and*

$$\lim_{n\to\infty} \mathbb{E}[|X_n - X|] = 0.$$

In particular, $\mathbb{E}[X_n] \to \mathbb{E}[X]$.

Proof. Recall that uniform integrability means

$$\lim_{K\to\infty} \sup_{n\geq 1} \mathbb{E}\big(|X_n| \, \mathbf{1}_{\{|X_n|>K\}}\big) = 0.$$

Fix $\varepsilon > 0$. Choose $K > 0$ such that

$$\sup_{n\geq 1} \mathbb{E}\big(|X_n| \, \mathbf{1}_{\{|X_n|>K\}}\big) < \varepsilon. \tag{2}$$

Step 1: X is integrable and $\mathbb{E}(|X| \, \mathbf{1}_{\{|X|>K\}}) \leq \varepsilon$. Since $X_n \to X$ a.s., we have $\mathbf{1}_{\{|X|>K\}} \leq \liminf_{n\to\infty} \mathbf{1}_{\{|X_n|>K\}}$ a.s.

Proof. Fix $K > 0$. Since $X_n \to X$ a.s., there exists a set $\Omega_0 \in \mathcal{F}$ with $\mathcal{P}(\Omega_0) = 1$ such that $X_n(\omega) \to X(\omega)$ for all $\omega \in \Omega_0$. Fix $\omega \in \Omega_0$.

Case 1: $|X(\omega)| \leq K$. Then $\mathbf{1}_{\{|X|>K\}}(\omega) = 0$, while $\liminf_{n\to\infty} \mathbf{1}_{\{|X_n|>K\}}(\omega) \geq 0$, hence

$$\mathbf{1}_{\{|X|>K\}}(\omega) \leq \liminf_{n\to\infty} \mathbf{1}_{\{|X_n|>K\}}(\omega).$$

Case 2: $|X(\omega)| > K$. Let $\varepsilon := \frac{|X(\omega)|-K}{2} > 0$. Since $X_n(\omega) \to X(\omega)$, there exists $N = N(\omega)$ such that for all $n \geq N$,

$$|X_n(\omega) - X(\omega)| < \varepsilon.$$

By the reverse triangle inequality,

$$|X_n(\omega)| \geq |X(\omega)| - |X_n(\omega) - X(\omega)| > |X(\omega)| - \varepsilon = \frac{|X(\omega)| + K}{2} > K,$$

so $\mathbf{1}_{\{|X_n|>K\}}(\omega) = 1$ for all $n \geq N$. Therefore,

$$\liminf_{n\to\infty} \mathbf{1}_{\{|X_n|>K\}}(\omega) = 1 = \mathbf{1}_{\{|X|>K\}}(\omega).$$

In both cases we have, for all $\omega \in \Omega_0$,

$$\mathbf{1}_{\{|X|>K\}}(\omega) \leq \liminf_{n\to\infty} \mathbf{1}_{\{|X_n|>K\}}(\omega).$$

Since $\mathcal{P}(\Omega_0) = 1$, the inequality holds almost surely. $\square$

Also $|X| \leq \liminf_{n\to\infty} |X_n|$ a.s. does not necessarily hold, but we can use Fatou on the nonnegative random variables $|X_n|\mathbf{1}_{\{|X_n|>K\}}$ after restricting to the set $\{|X| > K\}$ as follows: for a.s. ω with $|X(\omega)| > K$, we eventually have $|X_n(\omega)| > K$, hence

$$|X(\omega)|\, \mathbf{1}_{\{|X(\omega)|>K\}} \leq \liminf_{n\to\infty} |X_n(\omega)|\, \mathbf{1}_{\{|X_n(\omega)|>K\}}.$$

Applying Fatou's lemma yields

$$\mathbb{E}\big(|X|\, \mathbf{1}_{\{|X|>K\}}\big) \leq \liminf_{n\to\infty} \mathbb{E}\big(|X_n|\, \mathbf{1}_{\{|X_n|>K\}}\big) \leq \sup_{n\geq 1} \mathbb{E}\big(|X_n|\, \mathbf{1}_{\{|X_n|>K\}}\big) < \varepsilon.$$

In particular, $\mathbb{E}|X| < \infty$ since

$$\mathbb{E}|X| \leq K + \mathbb{E}\big(|X|\, \mathbf{1}_{\{|X|>K\}}\big) < K + \varepsilon.$$

Step 2: control the tails of $|X_n - X|$. For every n,

$$|X_n - X| \leq |X_n| + |X|,$$

so

$$\begin{aligned}\mathbb{E}\big(|X_n - X|\,\mathbf{1}_{\{|X_n|>K\}\cup\{|X|>K\}}\big) &\le \mathbb{E}\big(|X_n|\,\mathbf{1}_{\{|X_n|>K\}}\big) + \mathbb{E}\big(|X|\,\mathbf{1}_{\{|X|>K\}}\big)\\ &< \varepsilon + \varepsilon = 2\varepsilon,\end{aligned}$$

using (2) and Step 1.

Step 3: dominated convergence on the truncated part. Define

$$A_{n,K} := \{|X_n| \le K,\ |X| \le K\}.$$

On $A_{n,K}$ we have $|X_n - X| \le 2K$, and since $X_n \to X$ a.s.,

$$|X_n - X|\,\mathbf{1}_{A_{n,K}} \ \longrightarrow\ 0 \quad \text{a.s.}$$

Therefore, by the dominated convergence theorem,

$$\mathbb{E}\big(|X_n - X|\,\mathbf{1}_{A_{n,K}}\big) \longrightarrow 0 \qquad \text{as } n \to \infty.$$

Step 4: conclude L^1 convergence. Split

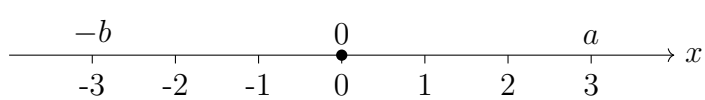
$$\mathbb{E}|X_n - X| = \mathbb{E}\big(|X_n - X|\,\mathbf{1}_{A_{n,K}}\big) + \mathbb{E}\Big(|X_n - X|\,\mathbf{1}_{A_{n,K}^c}\Big),$$

and note $A_{n,K}^c \subseteq \{|X_n| > K\} \cup \{|X| > K\}$. Hence, by Steps 2 and 3, for all sufficiently large n,

$$\mathbb{E}|X_n - X| \le \mathbb{E}\big(|X_n - X|\,\mathbf{1}_{A_{n,K}}\big) + 2\varepsilon < 3\varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, we conclude $\mathbb{E}|X_n - X| \to 0$. Finally, $\mathbb{E}[X_n] \to \mathbb{E}[X]$ follows from $|\mathbb{E}[X_n] - \mathbb{E}[X]| \le \mathbb{E}|X_n - X| \to 0$. $\square$

Gambler's Ruin



$$\boxed{\mathbb{P}(\tau_a < \tau_{-b}) = \frac{b}{a+b}, \qquad \mathbb{E}(\tau_a \wedge \tau_{-b}) = ab}$$

Remark 14 (Gambler's ruin without martingales: difference equations). Fix $a, b \in \mathbb{N}$ and let $(S_n)_{n\geq 0}$ be the simple random walk on $\mathbb{Z}$ with $S_0 = 0$ and steps ± 1 each with probability $1/2$. Define the boundary hitting times

$$\tau_a := \inf\{n \geq 0 : S_n = a\}, \qquad \tau_{-b} := \inf\{n \geq 0 : S_n = -b\}, \qquad T := \tau_a \wedge \tau_{-b}.$$

We derive $\mathbb{P}(\tau_a < \tau_{-b}) = \frac{b}{a+b}$ and $\mathbb{E}(T) = ab$ using only the Markov property and elementary recursions.

(i) Hitting probability. For each $x \in \{-b, -b+1, \ldots, a\}$ define

$$u(x) := \mathbb{P}_x(\tau_a < \tau_{-b}),$$

where $\mathbb{P}_x$ denotes the law of the walk started at $S_0 = x$. Clearly $u(-b) = 0$ and $u(a) = 1$. For interior points $-b < x < a$, condition on the first step: from x the walk moves to $x+1$ or $x-1$ with probability $1/2$, hence

$$u(x) = \frac{1}{2}\, u(x+1) + \frac{1}{2}\, u(x-1), \qquad -b < x < a.$$

Equivalently,

$$u(x+1) - u(x) = u(x) - u(x-1),$$

so the increments of u are constant and u must be affine: $u(x) = \alpha x + \beta$. Imposing the boundary conditions yields

$$0 = u(-b) = -\alpha b + \beta, \qquad 1 = u(a) = \alpha a + \beta,$$

so $\alpha = \frac{1}{a+b}$ and $\beta = \frac{b}{a+b}$. Therefore

$$\mathbb{P}(\tau_a < \tau_{-b}) = u(0) = \frac{b}{a+b}.$$

(ii) Expected exit time. For $x \in \{-b, \dots, a\}$ define the expected time to hit either boundary:

$$v(x) := \mathbb{E}_x(T).$$

We have boundary conditions $v(-b) = v(a) = 0$. For interior points $-b < x < a$, condition on the first step and use the Markov property: one step is taken immediately, and then the process restarts from $x \pm 1$, so

$$v(x) = 1 + \frac{1}{2}\, v(x+1) + \frac{1}{2}\, v(x-1), \qquad -b < x < a.$$

Rearranging gives the second-difference equation

$$v(x+1) - 2v(x) + v(x-1) = -2.$$

A standard way to solve this is to note that a quadratic has constant second difference. Indeed, try $v(x) = Ax^2 + Bx + C$. Then

$$v(x+1) - 2v(x) + v(x-1) = 2A,$$

so we need $2A = -2$, hence $A = -1$. Thus $v(x) = -x^2 + Bx + C$. Impose $v(a) = 0$ and $v(-b) = 0$:

$$0 = v(a) = -a^2 + Ba + C, \qquad 0 = v(-b) = -b^2 - Bb + C.$$

Subtracting gives $Ba + C = a^2$ and $-Bb + C = b^2$, hence $B = a - b$ and $C = ab$. Therefore

$$v(x) = -x^2 + (a-b)x + ab.$$

In particular, starting from 0,

$$\mathbb{E}(T) = v(0) = ab.$$

Interpretation. The function u is the unique discrete harmonic function on $\{-b, \dots, a\}$ with boundary values 0 and 1, hence it must be linear. The function v solves the discrete Poisson equation with forcing -2 and zero boundary values, hence it must be quadratic,

yielding $\mathbb{E}(T) = ab$.

Definition (Martingale). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(\mathcal{F}_t)_{t\geq 0}$ a filtration. A stochastic process $(X_t)_{t\geq 0}$ is a *martingale* with respect to $(\mathcal{F}_t)$ if, for all $t \geq 0$,

1. $\mathbb{E}|X_t| < \infty$ (integrable),
2. X_t is $\mathcal{F}_t$–measurable (adapted),
3. $\mathbb{E}[X_{t+1} \mid \mathcal{F}_t] = X_t$ almost surely (martingale property).

Explanation (Adaptedness / Measurability). Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and $(\mathcal{F}_t)_{t\geq 0}$ a filtration. For a random variable $X_t : \Omega \to \mathbb{R}$, the statement

$$X_t \text{ is } \mathcal{F}_t\text{–measurable}$$

means that

$$\{\, \omega \in \Omega : X_t(\omega) \in B \,\} \in \mathcal{F}_t \quad \text{for every Borel set } B \subset \mathbb{R}.$$

Equivalently,

$$\{X_t \leq a\} \in \mathcal{F}_t \quad \text{for all } a \in \mathbb{R}.$$

This condition ensures that the value of X_t is completely determined by the information available up to time t.

Remark 15 (Gambler's ruin: meaning and derivation of the formulas). Fix integers $a, b \in \mathbb{N}$ and consider the one–dimensional simple random walk $(S_n)_{n\geq 0}$ with $S_0 = 0$ and increments $X_n \in \{\pm 1\}$ i.i.d. with mean 0. Define the *hitting times*

$$\tau_a := \inf\{n \geq 0 : S_n = a\}, \qquad \tau_{-b} := \inf\{n \geq 0 : S_n = -b\},$$

and the *exit time* from the interval $[-b, a]$,

$$T := \tau_a \wedge \tau_{-b} = \min\{\tau_a, \tau_{-b}\}.$$

Thus T is the first time the walk hits either boundary a or $-b$.

(i) Hitting probability $\mathbb{P}(\tau_a < \tau_{-b})$. Since $(S_n)_{n\geq 0}$ is a martingale, we first apply optional stopping to the *bounded* stopping time $T \wedge N$ (for fixed $N \in \mathbb{N}$), to get

$$\mathbb{E}(S_{T\wedge N}) = \mathbb{E}(S_0) = 0, \qquad \forall\, N \in \mathbb{N}.$$

Next, note that $T < \infty$ a.s. (the walk exits the finite interval $[-b, a]$ almost surely), so $S_{T\wedge N} \to S_T$ a.s. as $N \to \infty$. Moreover, for every ω and every N,

$$\left|S_{T\wedge N}(\omega)\right| \leq \max\{a, b\},$$

since the walk stays in $[-b, a]$ up to time T and at time T it equals either a or $-b$. Therefore, by dominated convergence,

$$\mathbb{E}(S_{T\wedge N}) \xrightarrow[N\to\infty]{} \mathbb{E}(S_T),$$

and hence

$$\mathbb{E}(S_T) = 0.$$

But $S_T \in \{a, -b\}$ and

$$S_T = a\,\mathbf{1}_{\{\tau_a<\tau_{-b}\}} + (-b)\,\mathbf{1}_{\{\tau_{-b}<\tau_a\}}.$$

Let $p := \mathbb{P}(\tau_a < \tau_{-b})$. Then $\mathbb{P}(\tau_{-b} < \tau_a) = 1 - p$, and

$$0 = \mathbb{E}(S_T) = a\,p - b\,(1-p).$$

Solving for p yields

$$\mathbb{P}(\tau_a < \tau_{-b}) = \frac{b}{a+b}, \qquad \mathbb{P}(\tau_{-b} < \tau_a) = \frac{a}{a+b}.$$

(ii) Expected exit time $\mathbb{E}(\tau_a \wedge \tau_{-b})$. Consider the process

$$M_n := S_n^2 - n.$$

Since $\mathbb{E}(X_{n+1} \mid \mathcal{A}_n) = 0$ and $\mathbb{E}(X_{n+1}^2 \mid \mathcal{A}_n) = 1$, one checks that (M_n) is a martingale (the "square minus time" martingale; see next remark for full proof). For each fixed $N \in \mathbb{N}$, the stopping time $T \wedge N$ is bounded, so optional stopping yields

$$\mathbb{E}(M_{T\wedge N}) = \mathbb{E}(M_0) = 0, \qquad \forall\, N \in \mathbb{N}.$$

Expanding $M_{T\wedge N}$ gives

$$\mathbb{E}\big(S_{T\wedge N}^2\big) = \mathbb{E}(T \wedge N).$$

Next, since $T < \infty$ a.s., we have $S_{T\wedge N}^2 \to S_T^2$ a.s. and $T \wedge N \to T$ a.s. as $N \to \infty$. Moreover,

$$0 \le T \wedge N \le T \quad \text{and} \quad S_{T\wedge N}^2 \le \max\{a^2, b^2\} \quad \text{a.s.}$$

Hence $S_{T\wedge N}^2$ is dominated by the integrable constant $\max\{a^2, b^2\}$[1] , so by dominated convergence

$$\mathbb{E}\big(S_{T\wedge N}^2\big) \xrightarrow[N\to\infty]{} \mathbb{E}(S_T^2).$$

[1]Because dominated convergence is a theorem about *limits*: it allows one to interchange "limit" and "expectation". Concretely, if

$$X_N \to X \quad \text{a.s.} \qquad \text{and} \qquad |X_N| \le Y \text{ a.s. for all } N \text{ for some } Y \in L^1,$$

then

$$\mathbb{E}[X_N] \xrightarrow[N\to\infty]{} \mathbb{E}[X].$$

So if you want to conclude

$$\mathbb{E}(T \wedge N) \xrightarrow[N\to\infty]{} \mathbb{E}(T),$$

you must verify that the random variables you are taking expectations of actually converge to T, namely

$$T \wedge N \to T \quad \text{a.s.}$$

This holds on the event $\{T < \infty\}$ because once $N \ge T(\omega)$, we have

$$(T \wedge N)(\omega) = T(\omega),$$

so the sequence $(T \wedge N)(\omega)$ is eventually constant (and hence convergent) for each such ω. /

Also, since $S_T \in \{a, -b\}$, we have $S_T^2 \leq \max\{a^2, b^2\}$ a.s., and therefore

$$T = S_T^2 - M_T \leq \max\{a^2, b^2\} - M_T.$$

In particular, T is integrable (indeed $\mathbb{E}(T) < \infty$), and thus $(T \wedge N)$ is dominated by the integrable random variable T. By dominated convergence again,

$$\mathbb{E}(T \wedge N) \xrightarrow[N \to \infty]{} \mathbb{E}(T).$$

Passing to the limit in $\mathbb{E}(S_{T \wedge N}^2) = \mathbb{E}(T \wedge N)$ gives

$$\mathbb{E}(S_T^2) = \mathbb{E}(T).$$

Now $S_T \in \{a, -b\}$, so

$$S_T^2 = a^2 \, \mathbf{1}_{\{\tau_a < \tau_{-b}\}} + b^2 \, \mathbf{1}_{\{\tau_{-b} < \tau_a\}}.$$

Using $p = \mathbb{P}(\tau_a < \tau_{-b}) = \frac{b}{a+b}$ from part (i),

$$\mathbb{E}(S_T^2) = a^2 p + b^2(1-p) = a^2 \frac{b}{a+b} + b^2 \frac{a}{a+b} = \frac{ab(a+b)}{a+b} = ab.$$

Therefore,

$$\mathbb{E}(\tau_a \wedge \tau_{-b}) = \mathbb{E}(T) = ab.$$

Interpretation. The probability formula reflects linearity of harmonic functions for a fair walk: starting at 0 inside $[-b, a]$, the chance of exiting at a is proportional to the distance to the opposite boundary $-b$. The expected time formula shows that typical exit time scales quadratically in the interval length: larger barriers take longer to reach, with $\mathbb{E}(T) = ab$ capturing the product of the two distances from the starting point to the boundaries.

Theorem 4 (Dominated Convergence Theorem)**.** *Let $(X_n)_{n \geq 1}$ be integrable random variables and suppose that*

$$X_n \to X \quad a.s.$$

If there exists an integrable random variable Y such that

$$|X_n| \leq Y \quad a.s. \text{ for all } n,$$

then X is integrable and

$$\mathbb{E}[X_n] \to \mathbb{E}[X].$$

Proposition 1. *Let $(X_n)_{n\geq 1}$ be i.i.d. random variables with $\mathbb{E}(X_1) = 0$ and $\mathbb{E}(X_1^2) = 1$. Let $S_n = \sum_{k=1}^n X_k$ and let $\mathcal{A}_n = \sigma(X_1, \ldots, X_n)$. Define*

$$M_n := S_n^2 - n.$$

Then $(M_n)_{n\geq 0}$ is a martingale with respect to $(\mathcal{A}_n)_{n\geq 0}$.

Proof. First note that S_n is $\mathcal{A}_n$–measurable, hence $M_n = S_n^2 - n$ is $\mathcal{A}_n$–measurable. Moreover, since $\mathbb{E}(X_1^2) = 1 < \infty$, we have $\mathbb{E}(S_n^2) < \infty$, so $\mathbb{E}(|M_n|) < \infty$. Thus M_n is integrable.

We now verify the martingale property:

$$\mathbb{E}(M_{n+1} \mid \mathcal{A}_n) = M_n.$$

Observe that

$$S_{n+1} = S_n + X_{n+1},$$

so

$$S_{n+1}^2 = (S_n + X_{n+1})^2 = S_n^2 + 2S_nX_{n+1} + X_{n+1}^2.$$

Therefore,

$$M_{n+1} = S_{n+1}^2 - (n+1) = S_n^2 + 2S_nX_{n+1} + X_{n+1}^2 - (n+1).$$

Taking conditional expectation given $\mathcal{A}_n$,

$$\mathbb{E}(M_{n+1} \mid \mathcal{A}_n) = S_n^2 + 2S_n\,\mathbb{E}(X_{n+1} \mid \mathcal{A}_n) + \mathbb{E}(X_{n+1}^2 \mid \mathcal{A}_n) - (n+1).$$

Since X_{n+1} is independent of $\mathcal{A}_n$ and $\mathbb{E}(X_{n+1}) = 0$, we have

$$\mathbb{E}(X_{n+1} \mid \mathcal{A}_n) = 0.$$

Similarly, independence and $\mathbb{E}(X_{n+1}^2) = 1$ imply

$$\mathbb{E}(X_{n+1}^2 \mid \mathcal{A}_n) = 1.$$

Hence

$$\mathbb{E}(M_{n+1} \mid \mathcal{A}_n) = S_n^2 + 0 + 1 - (n+1) = S_n^2 - n = M_n.$$

Thus (M_n) satisfies

$$\mathbb{E}(M_{n+1} \mid \mathcal{A}_n) = M_n,$$

so it is a martingale. $\square$

Professor's proof of gambler's ruin

We consider the simple symmetric random walk

$$S_n := \sum_{k=1}^{n} X_k, \qquad S_0 = 0,$$

where $(X_k)_{k\geq 1}$ are i.i.d. with $\mathbb{P}(X_k = 1) = \mathbb{P}(X_k = -1) = \frac{1}{2}$. Let

$$\tau_a := \inf\{n \geq 0 : S_n = a\}, \qquad \tau_{-b} := \inf\{n \geq 0 : S_n = -b\},$$

for integers $a, b > 0$.

Step 1: Truncation. Define the bounded stopping time

$$T_N := (\tau_a \wedge \tau_{-b}) \wedge N.$$

Since $T_N \le N$, the optional stopping corollary for bounded stopping times applies:

$$\mathbb{E}[S_{T_N}] = 0, \qquad \mathbb{E}[S_{T_N}^2] = \mathbb{E}[T_N].$$

Step 2: Probability of ruin. Decompose S_{T_N} according to the event that occurs first. On $\{\tau_a < \tau_{-b},\, \tau_a \le N\}$ we have $S_{T_N} = a$; on $\{\tau_{-b} < \tau_a,\, \tau_{-b} \le N\}$ we have $S_{T_N} = -b$; and on $\{T_N = N\} = \{\tau_a \wedge \tau_{-b} > N\}$ we have $S_{T_N} = S_N$ with $-b < S_N < a$. Thus

$$0 = a\,\mathbb{P}(\tau_a < \tau_{-b},\, \tau_a \le N) - b\,\mathbb{P}(\tau_{-b} < \tau_a,\, \tau_{-b} \le N) + \mathbb{E}\big[S_N\,\mathbf{1}_{\{\tau_a \wedge \tau_{-b} > N\}}\big]\,.$$

Since $|S_N| < \max\{a, b\}$ on $\{\tau_a \wedge \tau_{-b} > N\}$,

$$\big|\mathbb{E}\big[S_N\,\mathbf{1}_{\{\tau_a \wedge \tau_{-b} > N\}}\big]\big| \le \max\{a, b\}\,\mathbb{P}(\tau_a \wedge \tau_{-b} > N).$$

Letting $N \to \infty$ and using $\mathbb{P}(\tau_a \wedge \tau_{-b} > N) \to 0$, we obtain

$$a\,\mathbb{P}(\tau_a < \tau_{-b}) = b\,\mathbb{P}(\tau_{-b} < \tau_a).$$

Since these two events partition the sample space,

$$\mathbb{P}(\tau_a < \tau_{-b}) = \frac{b}{a+b}, \qquad \mathbb{P}(\tau_{-b} < \tau_a) = \frac{a}{a+b}.$$

Step 3: Expected duration of the game. Using $\mathbb{E}[S_{T_N}^2] = \mathbb{E}[T_N]$, we similarly decompose $S_{T_N}^2$:

$$\mathbb{E}[T_N] = a^2\,\mathbb{P}(\tau_a < \tau_{-b},\, \tau_a \le N) + b^2\,\mathbb{P}(\tau_{-b} < \tau_a,\, \tau_{-b} \le N) + \mathbb{E}\big[S_N^2\,\mathbf{1}_{\{\tau_a \wedge \tau_{-b} > N\}}\big]\,.$$

On $\{\tau_a \wedge \tau_{-b} > N\}$ we have $S_N^2 \le \max\{a, b\}^2$, so the last term is bounded by

$$\max\{a, b\}^2\,\mathbb{P}(\tau_a \wedge \tau_{-b} > N) \xrightarrow[N \to \infty]{} 0.$$

Letting $N \to \infty$ and substituting the hitting probabilities yields

$$\mathbb{E}[\tau_a \wedge \tau_{-b}] = a^2 \frac{b}{a+b} + b^2 \frac{a}{a+b} = ab.$$

This completes the proof of the gambler's ruin formulas using truncation and optional stopping.

Reflection Principle

Reflection Principle

The number of paths from a to b in N steps that hit 0 equals the number of paths from $-a$ to b in N steps.

Reflection Principle (multiple hits allowed)

Fix $a > 0$, $b \in \mathbb{Z}$, and $N \in \mathbb{N}$. Let $(S_n)_{0 \le n \le N}$ be a nearest–neighbor path with $S_0 = a$ and $S_N = b$, and define the *first* hitting time of 0 by

$$\tau_0 := \inf\{n \ge 0 : \ S_n = 0\}.$$

Consider the set

$$\mathcal{P} := \{(S_0, \dots, S_N) : S_0 = a, \ S_N = b, \ \tau_0 \le N\},$$

i.e. paths that hit 0 at least once (possibly many times). Define the reflected path $\widetilde{S}$ by reflecting only up to the first hit:

$$\widetilde{S}_n := \begin{cases} -S_n, & 0 \le n \le \tau_0, \\ S_n, & \tau_0 \le n \le N. \end{cases}$$

Then $\widetilde{S}_0 = -a$, $\widetilde{S}_N = b$, and the map $S \mapsto \widetilde{S}$ is a bijection between

$$\{\text{paths from } a \text{ to } b \text{ in } N \text{ steps that hit } 0\} \quad \text{and} \quad \{\text{paths from } -a \text{ to } b \text{ in } N \text{ steps}\}.$$

In particular,

$$\#\{a \to b \text{ paths in } N \text{ steps that hit } 0\} = \#\{-a \to b \text{ paths in } N \text{ steps}\}.$$

Reflection Principle (arbitrary nonzero start)

Fix $a \in \mathbb{Z} \setminus \{0\}$, $b \in \mathbb{Z}$, and $N \in \mathbb{N}$. Let $(S_n)_{0 \le n \le N}$ be a nearest–neighbor path with $S_0 = a$ and $S_N = b$, and define

$$\tau_0 := \inf\{n \ge 0 : \ S_n = 0\}.$$

Define the reflected path $\widetilde{S}$ by reflecting only up to the first hit:

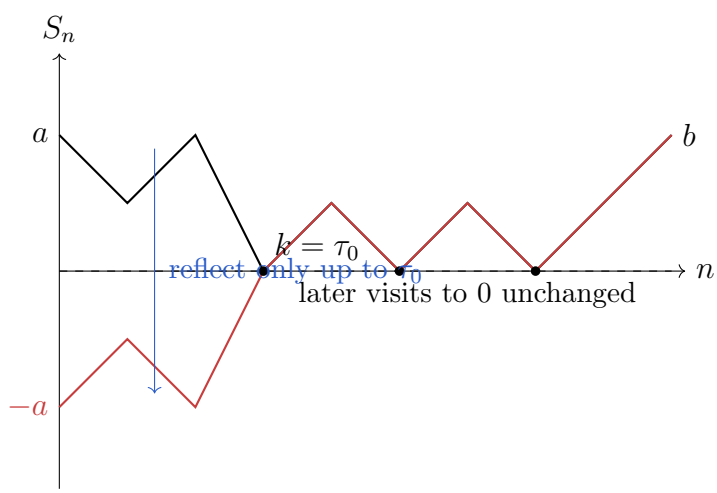
$$\widetilde{S}_n := \begin{cases} -S_n, & 0 \le n \le \tau_0, \\ S_n, & \tau_0 \le n \le N. \end{cases}$$

Then the map $S \mapsto \widetilde{S}$ is a bijection between

$$\{\text{paths from } a \text{ to } b \text{ in } N \text{ steps that hit } 0\} \quad \text{and} \quad \{\text{paths from } -a \text{ to } b \text{ in } N \text{ steps}\}.$$

In particular,

$$\#\{a \to b \text{ paths in } N \text{ steps that hit } 0\} = \#\{-a \to b \text{ paths in } N \text{ steps}\}.$$



Ballot Theorem

Theorem (Ballot Theorem)

The number of paths from $(0,0)$ to (N,b) that never return to 0 equals

$$\frac{b}{N} W_N(0,b)$$

Recurrence and Arcsine Law

Last zero before time $2N$ and the arcsine law. Define the last visit to 0 up to time $2N$ by

$$L := \max\{0 \le k \le 2N : \ S_k = 0\}.$$

Then for $k = 0, 1, \ldots, N$ we have the exact event identity

$$\{L = 2k\} = \{S_{2k} = 0\} \ \cap \ \{S_j \neq 0 \text{ for all } j = 2k+1, \ldots, 2N\}.$$

Equivalently, if we define the *first return to* 0 *after time* $2k$ by

$$T_0^{(2k)} := \inf\{m \ge 1 : \ S_{2k+m} = 0\},$$

then

$$\{L = 2k\} = \{S_{2k} = 0\} \cap \{T_0^{(2k)} > 2N - 2k\}.$$

In particular, the condition "no return after $2k$" implies $S_{2N} \neq 0$ unless $k = N$.

Arcsine law for the last zero

As $N \to \infty$,

$$\mathbb{P}\left(\frac{L}{2N} \le x\right) \longrightarrow \frac{2}{\pi} \arcsin \sqrt{x}, \qquad x \in [0,1].$$

Conclusion

The one-dimensional simple random walk is null recurrent:

$$\mathbb{E}[T_0] = \infty,$$

where $T_0 := \inf\{n \geq 1 : \ S_n = 0\}$ is the first return time to 0.

Conclusion

SRW is null recurrent:

$$\mathbb{E}(\tau_0) = \infty.$$

Lecture Notes: (Jan. 27, 2026)

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space. Let $(X_n)_{n\geq 1}$ be a sequence of i.i.d. random variables, and let S_0 be independent of $(X_n)_{n\geq 1}$.

$$S_n = S_0 + \sum_{j=1}^{n} X_j, \qquad n = 1, 2, \ldots$$

We call $(S_n)_{n\geq 0}$ a *random walk*.

Simple Random Walk (SRW)

Special Case: Simple Random Walk

$$X_n \in \{\pm e_1, \ldots, \pm e_d\}, \qquad \mathbb{P}(X_n = \pm e_k) = \frac{1}{2d}.$$

Typically we start at the origin:

$$\mathbb{P}(S_0 = 0) = 1.$$

SRW in Finite Time

Fix $N \in \mathbb{N}$. Let $\Omega_N = \{-1, 1\}^N$ with

$$X_n(\omega) = \omega_n, \qquad \mathbb{P}(A) = \frac{|A|}{2^N}.$$

Exercise 1

Show that $X_1, \ldots, X_N$ are i.i.d. and

$$\mathbb{P}(X_i = \pm 1) = \frac{1}{2}.$$

Solution (Exercise 1)

Recall $\Omega_N = \{-1, 1\}^N$ and $\mathbb{P}(A) = |A|/2^N$. For each i define $X_i(\omega) = \omega_i$.

Step 1 (marginals). Fix $i \in \{1, \dots, N\}$. The event $\{X_i = 1\}$ is the set of sequences in Ω_N whose ith coordinate equals 1. Exactly half of the 2^N sequences have $\omega_i = 1$, hence

$$\mathbb{P}(X_i = 1) = \frac{2^{N-1}}{2^N} = \frac{1}{2}, \qquad \mathbb{P}(X_i = -1) = 1 - \mathbb{P}(X_i = 1) = \frac{1}{2}.$$

Step 2 (independence). Let $1 \le k \le N$ and pick distinct indices $i_1, \dots, i_k$ with prescribed values $\varepsilon_1, \dots, \varepsilon_k \in \{-1, 1\}$. Then

$$\{X_{i_1} = \varepsilon_1, \dots, X_{i_k} = \varepsilon_k\}$$

fixes k coordinates and leaves the remaining $N - k$ coordinates free, so the set has cardinality 2^{N-k}. Therefore

$$\mathbb{P}(X_{i_1} = \varepsilon_1, \dots, X_{i_k} = \varepsilon_k) = \frac{2^{N-k}}{2^N} = 2^{-k} = \prod_{j=1}^{k} \mathbb{P}(X_{i_j} = \varepsilon_j),$$

which proves independence.

Step 3 (identical distribution). The marginal computation in Step 1 does not depend on i, so all X_i have the same distribution. Hence $X_1, \dots, X_N$ are i.i.d.

Exercise 2

Show that SRW has stationary independent increments: for any $0 \le k_0 < k_1 < \cdots < k_m \le N$, the increments

$$S_{k_1} - S_{k_0},\ S_{k_2} - S_{k_1},\ \dots,\ S_{k_m} - S_{k_{m-1}}$$

are independent, and for $0 \le k < m \le N$,

$$S_m - S_k \overset{d}{=} S_{m-k}.$$

Solution (Exercise 2)

Recall $S_n = \sum_{j=1}^{n} X_j$ (with $S_0 = 0$), where $X_1, \ldots, X_N$ are i.i.d.

Step 1 (write increments as disjoint block sums). For any $0 \le a < b \le N$,

$$S_b - S_a = \sum_{j=a+1}^{b} X_j.$$

Hence for $0 \le k_0 < k_1 < \cdots < k_m \le N$,

$$S_{k_r} - S_{k_{r-1}} = \sum_{j=k_{r-1}+1}^{k_r} X_j, \qquad r = 1, \ldots, m.$$

Each increment is a sum of a *disjoint block* of the X_j's, and these blocks are pairwise disjoint across r.

Step 2 (independent increments). Because the X_j's are independent, any two sums that depend on disjoint sets of indices are independent. Since the blocks

$$\{k_{r-1} + 1, \ldots, k_r\}, \quad r = 1, \ldots, m$$

are disjoint, it follows that the increments

$$S_{k_1} - S_{k_0},\ S_{k_2} - S_{k_1},\ \ldots,\ S_{k_m} - S_{k_{m-1}}$$

are independent.

Step 3 (stationarity of increments). Fix $0 \le k < m \le N$. Then

$$S_m - S_k = \sum_{j=k+1}^{m} X_j.$$

This is a sum of $(m - k)$ i.i.d. $\{\pm 1\}$–valued variables. By identical distribution, the block $(X_{k+1}, \ldots, X_m)$ has the same joint distribution as $(X_1, \ldots, X_{m-k})$. Therefore

$$S_m - S_k = \sum_{j=k+1}^{m} X_j \overset{d}{=} \sum_{j=1}^{m-k} X_j = S_{m-k}.$$

Conclusion. The SRW has independent increments (Step 2) and stationary increments (Step 3).

Moments and Distribution

$$\mathbb{E}(S_n) = 0, \qquad \mathrm{Var}(S_n) = n.$$

$$\mathbb{P}(S_n = x) = \binom{n}{\frac{n+x}{2}} 2^{-n}, \qquad x \equiv n \pmod 2$$

$$\mathbb{P}(S_n = x) = \mathbb{P}(S_n = -x)$$

Stopping Times

Let $\mathcal{A}_n = \sigma(X_1, \ldots, X_n)$.

Definition (Stopping Time)

A random variable $T : \Omega \to \{0, 1, \ldots, N\} \cup \{\infty\}$ is a stopping time if

$$\{T = n\} \in \mathcal{A}_n.$$

$$\tau_a = \inf\{n \geq 0 : S_n = a\} \quad \text{is a stopping time.}$$

Optional Stopping

Theorem

For any predictable strategy (V_k),

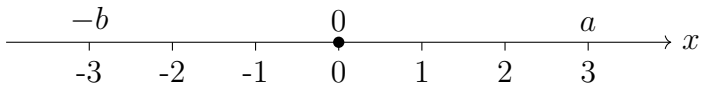
$$\mathbb{E}\left(\sum_{k=1}^{N} V_k X_k\right) = 0.$$

Corollary

If T is a bounded stopping time, then

$$\mathbb{E}(S_T) = 0, \qquad \mathrm{Var}(S_T) = \mathbb{E}(T).$$

Gambler's Ruin



$$\mathbb{P}(\tau_a < \tau_{-b}) = \frac{b}{a+b}, \qquad \mathbb{E}(\tau_a \wedge \tau_{-b}) = ab$$

Random walks in infinite time

Feb. 3, 2026

Wald's equations

Let $(S, \mathcal{S})$ be a "nice" measurable space in the sense that there is a one–to–one measurable map from S to $\mathbb{R}$ (with $\mathbb{R}$ endowed with its Borel σ–algebra) such that the inverse is also measurable.

(**Theorem 2.1.22** in Durrett: If S is a Borel subset of a complete separable metric space M, and $\mathcal{S}$ is the collection of Borel subsets of S, then $(S, \mathcal{S})$ is nice.)

Let

$$\Omega := S^{\mathbb{N}} = \{(\omega_1, \omega_2, \ldots) : \omega_i \in S\}, \qquad \mathcal{F} := \mathcal{S} \otimes \mathcal{S} \otimes \cdots .$$

Let

$$\mathbb{P} := \mu \times \mu \times \cdots ,$$

where μ is a probability measure on $(S, \mathcal{S})$.

Define

$$X_n(\omega) = \omega_n,$$

so that $\mathbb{P}(X_n \in B) = \mu(B)$ and $(X_n)_{n\geq 1}$ are independent by construction. If $\mu_n = \mu$ for all n, then $(X_n)_{n\geq 1}$ is an i.i.d. sequence.

Define the filtrations

$$\mathcal{F}_n := \sigma(X_1, \ldots, X_n), \qquad \mathcal{F}'_n := \sigma(X_{n+1}, X_{n+2}, \ldots), \qquad n = 0, 1, 2, \ldots$$

Example: Random walk in $\mathbb{R}^d$

We shall be mostly interested in random walks in $\mathbb{R}^d$:

$$(S, \mathcal{S}) = (\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d)), \qquad (X_n)_{n\geq 1} \text{ i.i.d.},$$

with $S_0 = 0$ and

$$S_n := \sum_{i=1}^{n} X_i.$$

We shall discuss two 0–1 laws and some of their applications: Kolmogorov's 0–1 law and Hewitt–Savage 0–1 law.

Permutable events and exchangeable σ–algebra

Assume now that $\mu_n = \mu$ for all n, so $(X_n)_{n\geq 1}$ are i.i.d.

A map $\pi : \mathbb{N} \to \mathbb{N}$ is a *finite permutation* if $\pi(i) \neq i$ only for finitely many $i \in \mathbb{N}$.

Define

$$(\pi\omega)_i := \omega_{\pi(i)}, \qquad i \in \mathbb{N}.$$

Then

$$S_n(\omega) = S_n(\pi\omega) \quad \text{for all } n \geq 1.$$

An event $A \in \mathcal{F}$ is *permutable* if

$$\pi^{-1}A := \{\omega : \pi\omega \in A\} = A \quad \text{for all finite permutations } \pi.$$

Examples

Suppose $S = \mathbb{R}$, $A \subset \mathbb{R}$, $B \in \mathcal{B}(\mathbb{R})$.

1. $$\{\omega : S_n(\omega) \in B \text{ i.o.}\} = \bigcap_{m\geq 1} \bigcup_{n\geq m} \{\omega : S_n(\omega) \in B\}.$$

2. $$\{\omega : \limsup_{n\to\infty} S_n(\omega) \in B\}$$

$$\{\omega : \limsup_{n\to\infty} S_n(\omega) \in B\} = \bigcap_{m\geq 1} \{\omega : \sup_{n\geq m} S_n(\omega) \in B\}$$

provided B is closed.

If you don't want to assume B closed, you can use outer approximations $B^{(1/k)} := \{x : \mathrm{dist}(x, B) < 1/k\}$ (open thickenings):

$$\{\omega : \limsup_{n\to\infty} S_n(\omega) \in B\} = \bigcap_{k\geq 1} \bigcap_{m\geq 1} \bigcup_{N\geq m} \{\omega : \sup_{m\leq n\leq N} S_n(\omega) \in B^{(1/k)}\}.$$

That last one is the closest analogue to your first-item "$\cap_m \cup_{n\geq m}$" structure, but it's necessarily a bit more elaborate because $\limsup S_n$ involves **suprema of tails**, not just membership infinitely often.

3. Tail events:

$$\mathcal{T} := \bigcap_{n\geq 0} \mathcal{F}'_n.$$

The collection of all permutable events forms a σ–algebra called the *exchangeable σ–algebra*:

$$\mathcal{E} := \{A \in \mathcal{F} : \pi^{-1}A = A \text{ for all finite permutations } \pi\}.$$

Note that $\mathcal{T} \subset \mathcal{E}$, but $\mathcal{E}$ is strictly larger than $\mathcal{T}$, as illustrated by (i).

Remark 16 (Two different "lim sup"s.). **(A)** lim sup **of a sequence of events.** Your first item is the lim sup of the events $A_n := \{S_n \in B\}$:

$$\{S_n \in B \text{ i.o.}\} = \limsup_{n\to\infty}\{S_n \in B\} := \bigcap_{m\geq 1} \bigcup_{n\geq m} \{S_n \in B\}.$$

Here lim sup is applied to the sequence of events $(A_n)_{n\geq 1}$.

(B) lim sup **of a sequence of real numbers.** Your second item is the lim sup of the real sequence $S_n(\omega)$, taken pointwise in ω:

$$\left\{\limsup_{n\to\infty} S_n \in B\right\}.$$

Here lim sup is applied to $(S_n(\omega))_{n\geq 1}$, producing the random variable

$$\limsup_{n\to\infty} S_n(\omega) = \lim_{m\to\infty} \sup_{n\geq m} S_n(\omega).$$

Why they are different (quick sanity check). Take $B = (0, \infty)$. Then $\{S_n \in B \text{ i.o.}\}$ means "$S_n > 0$ infinitely often." Whereas $\{\limsup_{n\to\infty} S_n \in B\}$ means "$\limsup_{n\to\infty} S_n > 0$." For example, if $S_n = 1/n$, then $S_n > 0$ for all n (so $\{S_n \in B \text{ i.o.}\}$ holds), but

$$\limsup_{n\to\infty} S_n = \lim_{n\to\infty} \frac{1}{n} = 0 \notin (0, \infty),$$

so $\{\limsup_{n\to\infty} S_n \in B\}$ fails.

Set-notation expansion for the second item. Using

$$\limsup_{n\to\infty} S_n = \inf_{m\geq 1} \sup_{n\geq m} S_n,$$

a standard events-only expansion (when B is closed) is

$$\left\{\limsup_{n\to\infty} S_n \in B\right\} = \bigcap_{m\geq 1} \left\{\sup_{n\geq m} S_n \in B\right\}.$$

Exercise 1

Which of the following events are in $\mathcal{T}$? In $\mathcal{E}$?

1. $$\{X_n \in B_n \text{ i.o.}\} = \bigcap_{m\geq 1} \bigcup_{n\geq m} \{\omega : X_n(\omega) \in B_n\}, \qquad B_n \in \mathcal{B}(\mathbb{R}^d).$$

2. $$\{\omega : \lim_{n\to\infty} S_n(\omega) \text{ exists}\}.$$

3. $$\{\omega : \limsup_{n\to\infty} S_n(\omega) > 0\}.$$

4.

$$\{\omega : \limsup_{n\to\infty} S_n(\omega)/c_n > x\}, \quad c_n \to \infty,\ x \in \mathbb{R}.$$

Theorem (Hewitt–Savage 0–1 law)

If $(X_n)_{n\geq 1}$ is a sequence of i.i.d. random variables and $A \in \mathcal{E}$, then

$$\mathbb{P}(A) \in \{0, 1\}.$$

(Kolmogorov 0–1 law: If $X_1, X_2, \ldots$ are independent and $A \in \mathcal{T}$, then $\mathbb{P}(A) \in \{0, 1\}$.)

Proof

We will show that A is independent of itself:

$$\mathbb{P}(A) = \mathbb{P}(A \cap A) = \mathbb{P}(A)\mathbb{P}(A),$$

which implies $\mathbb{P}(A) \in \{0, 1\}$.

Key fact: For all $A \in \mathcal{E}(X_1, X_2, \ldots)$, there exists $A_n \in \sigma(X_1, \ldots, X_n)$ such that

$$\mathbb{P}(A_n \triangle A) \to 0 \quad \text{as } n \to \infty.$$

Note also that

$$|\mathbb{P}(A) - \mathbb{P}(A_n)| \leq \mathbb{P}(A \triangle A_n) \to 0.$$

Exercise 2

Show that for all $A \in \mathcal{E}(X_1, X_2, \ldots)$, there exists $A_n \in \sigma(X_1, \ldots, X_n)$ such that

$$\mathbb{P}(A_n \triangle A) \to 0 \quad \text{as } n \to \infty.$$

Continuation of proof

Let

$$A_n := \{\omega : (\omega_1, \ldots, \omega_n) \in B_n\} \quad \text{for some } B_n \subset S^n.$$

Define a permutation π by

$$\pi(j) = \begin{cases} j+n, & j \in \{1, \ldots, n\}, \\ j-n, & j \in \{n+1, \ldots, 2n\}, \\ j, & j > 2n. \end{cases}$$

Then $\pi^2 = \text{id}$.

Since coordinates are i.i.d.,

$$\mathbb{P}(\omega : \omega \in A_n \cap A) = \mathbb{P}(\omega : \pi\omega \in A_n \cap A).$$

Because A is permutable,

$$\{\omega : \pi\omega \in A_n \cap A\} = \{\omega : \omega \in A'_n \cap A\},$$

where

$$A'_n := \{\omega : (\omega_{n+1}, \ldots, \omega_{2n}) \in B_n\}.$$

Thus,

$$\mathbb{P}(A_n \cap A) = \mathbb{P}(A'_n \cap A).$$

One checks that

$$\mathbb{P}(A'_n \triangle A) \leq \mathbb{P}(A'_n \triangle A_n) + \mathbb{P}(A_n \triangle A) \to 0.$$

Hence,

$$\mathbb{P}(A) = \lim_{n\to\infty} \mathbb{P}(A_n \cap A'_n) = \lim_{n\to\infty} \mathbb{P}(A_n)\mathbb{P}(A'_n) = (\mathbb{P}(A))^2.$$

Exercise 3

Let $X_1, X_2, \ldots \in \mathbb{R}$ be independent and let

$$\mathcal{T} = \bigcap_{n \geq 1} \sigma(X_n, X_{n+1}, \ldots).$$

Show that if a random variable X is $\mathcal{T}$–measurable, then it must be a constant, i.e. there exists c such that $\mathbb{P}(X = c) = 1$. Does the same hold in $\mathbb{R}^d$, $d > 1$?

Exercise 4 (Bernoulli percolation)

Consider Bernoulli percolation on $\mathbb{Z}^2$. To each edge associate $X_e \in \{0, 1\}$, where $(X_e)_{e \in E}$ are independent and

$$\mathbb{P}(X_e = 1) = p = 1 - \mathbb{P}(X_e = 0).$$

Let C_∞ be the event that there is an infinite cluster (an infinite connected subgraph). Show that

$$\mathbb{P}(C_\infty) \in \{0, 1\}.$$

Application to random walks

Theorem

For a random walk on $\mathbb{R}$ there are four possibilities, one of which has probability 1:

1. $S_n = 0$ for all n;
2. $S_n \to +\infty$ as $n \to \infty$;
3. $S_n \to -\infty$ as $n \to \infty$;
4. $-\infty = \liminf S_n < \limsup S_n = +\infty$.

Proof Idea from Class

By Hewitt–Savage 0–1 law, $\limsup S_n : \Omega \to [-\infty, \infty]$ is constant. If $\limsup S_n = c \in \mathbb{R}$, then $S_n - X_1$ has the same distribution as S_n, so $\limsup S_n = c - X_1$, implying $X_1 = 0$ a.s.,

giving (i). Otherwise $\limsup S_n = \pm\infty$. The same reasoning applies to $\liminf S_n$.

Proof.

Step 1: $\limsup S_n$ and $\liminf S_n$ are invariant under finite permutations (Hewitt–Savage setup).

Let π be a permutation of $\mathbb{N}$ that fixes all but finitely many indices. Equivalently, there exists a finite set $F \subset \mathbb{N}$ such that $\pi(k) = k$ for all $k \notin F$. Let

$$M := \max F \quad (\text{take } M = 0 \text{ if } F = \varnothing).$$

Define a permuted increment sequence by

$$X'_k := X_{\pi(k)}, \qquad k \geq 1,$$

and let

$$S'_n := \sum_{k=1}^{n} X'_k = \sum_{k=1}^{n} X_{\pi(k)}$$

be the associated partial sums.

Claim: For every $n \geq M$, we have $S'_n = S_n$.

Proof of the claim. Fix $n \geq M$. We show that $\{\pi(1), \ldots, \pi(n)\} = \{1, \ldots, n\}$. First, for any $k \leq n$ we clearly have $\pi(k) \leq n$: indeed, if $\pi(k) > n$, then there exists $j := \pi(k) > n$ with $\pi^{-1}(j) = k \leq n$. But since $j > n \geq M$ and π fixes all indices $> M$, we have $\pi(j) = j$ and hence $\pi^{-1}(j) = j > n$, contradicting $\pi^{-1}(j) = k \leq n$. Thus $\pi(\{1, \ldots, n\}) \subseteq \{1, \ldots, n\}$. Because π is a bijection, this inclusion must be an equality:

$$\{\pi(1), \ldots, \pi(n)\} = \{1, \ldots, n\}.$$

Therefore, $\pi(1), \ldots, \pi(n)$ is merely a reordering of $1, \ldots, n$, and hence

$$S'_n = \sum_{k=1}^{n} X_{\pi(k)} = \sum_{j=1}^{n} X_j = S_n,$$

proving the claim. $\diamond$

From $S'_n = S_n$ for all $n \geq M$, it follows immediately that

$$\limsup_{n\to\infty} S'_n = \limsup_{n\to\infty} S_n, \qquad \liminf_{n\to\infty} S'_n = \liminf_{n\to\infty} S_n.$$

Hence for every $a \in \mathbb{R}$ the events

$$\{\limsup_{n\to\infty} S_n \leq a\}, \qquad \{\liminf_{n\to\infty} S_n \geq a\},$$

and likewise the events $\{\limsup S_n = +\infty\}$, $\{\limsup S_n = -\infty\}$, etc., are invariant under every finite permutation of $(X_1, X_2, \dots)$.

Since the increments are i.i.d., the Hewitt–Savage 0–1 law implies that each such permutation-invariant event has probability 0 or 1. In particular, for each $a \in \mathbb{R}$,

$$\mathbb{P}\Big(\limsup_{n\to\infty} S_n \leq a\Big) \in \{0,1\}, \qquad \mathbb{P}\Big(\liminf_{n\to\infty} S_n \geq a\Big) \in \{0,1\}.$$

This forces $\limsup_{n\to\infty} S_n$ and $\liminf_{n\to\infty} S_n$ to be almost surely constant as extended real random variables. That is, there exist constants $c_+, c_- \in [-\infty, \infty]$ such that

$$\limsup_{n\to\infty} S_n = c_+ \quad \text{a.s.}, \qquad \liminf_{n\to\infty} S_n = c_- \quad \text{a.s.}.$$

Step 2: If c_+ is finite, then $X_1 = 0$ a.s.

Assume $c_+ \in \mathbb{R}$. Now observe the identity

$$S_n - X_1 = \sum_{k=2}^{n} X_k.$$

The sequence $(S_n - X_1)_{n\geq 1}$ has the same distribution as $(S_{n-1})_{n\geq 1}$, hence the same distribution as $(S_n)_{n\geq 0}$ (since (X_k) are iid sequence and pushforward preserves equality in law). Therefore,

$$\limsup_{n\to\infty}(S_n - X_1) \stackrel{d}{=} \limsup_{n\to\infty} S_n.$$

But

$$\limsup_{n\to\infty}(S_n - X_1) = \limsup_{n\to\infty} S_n - X_1.$$

Hence almost surely,

$$c_+ = c_+ - X_1.$$

Therefore $X_1 = 0$ almost surely. Since the increments are i.i.d., we conclude

$$X_k = 0 \quad \text{a.s. for all } k.$$

Thus

$$S_n = 0 \quad \text{for all } n \text{ a.s.}$$

which is case (1).

Step 3: Therefore $c_+ \in \{\pm\infty\}$.

If the walk is not identically zero, the previous step shows c_+ cannot be finite. Hence

$$\limsup S_n = +\infty \quad \text{or} \quad \limsup S_n = -\infty \quad \text{a.s.}$$

The same argument applied to $\liminf S_n$ yields

$$\liminf S_n \in \{\pm\infty\} \quad \text{a.s.}$$

Step 4: Classification of possibilities.

We now consider all combinations of (c_-, c_+):

Case A: $c_+ = +\infty$ and $c_- = +\infty$.

Then $\lim S_n = +\infty$, giving (2).

Case B: $c_+ = -\infty$ and $c_- = -\infty$.

Then $\lim S_n = -\infty$, giving (3).

Case C: $c_+ = +\infty$ and $c_- = -\infty$.

Then

$$-\infty = \liminf S_n < \limsup S_n = +\infty,$$

which is (4).

Case D: $c_+ = -\infty$ and $c_- = +\infty$.

This is impossible since always $\liminf S_n \leq \limsup S_n$.

Thus exactly one of the four stated possibilities occurs almost surely.

□

Exercise 5

If μ is symmetric with respect to 0 and $\mathbb{P}(X_1 = 0) < 1$, then we are in case (iv).

Stopping times

Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and

$$\mathcal{F}_0 \subset \mathcal{F}_1 \subset \mathcal{F}_2 \subset \cdots \subset \mathcal{F}$$

be a filtration.

A random variable $T : \Omega \to \{0, 1, 2, \ldots\} \cup \{\infty\}$ is a *stopping time* if

$$\{T \leq n\} \in \mathcal{F}_n \quad \text{for all } n \geq 0.$$

Example

Let $(S_n)_{n\geq 0}$ be a random walk and $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$. Define

$$T := \inf\{n \geq 0 : S_n \in A\}.$$

Then

$$\{T = n\} = \{S_n \in A\} \cap \bigcap_{i<n} \{S_i \notin A\} \in \mathcal{F}_n.$$

Exercise 6

If S, T are stopping times relative to $(\mathcal{F}_n)$, then $S \wedge T$, $S \vee T$, and $S+T$ are also stopping times. Give an example showing that $T - S$ need not be a stopping time, even if $\mathbb{P}(T > S) = 1$.

σ–algebra $\mathcal{F}_T$

If T is a stopping time, define

$$\mathcal{F}_T := \{A \in \mathcal{F} : A \cap \{T \leq n\} \in \mathcal{F}_n \text{ for all } n\}.$$

Then $\mathcal{F}_T$ is a σ–algebra and T is $\mathcal{F}_T$–measurable.

Theorem 2

Let $(X_n)_{n\geq 1}$ be i.i.d., $\mathcal{F}_n = \sigma(X_1, \ldots, X_n)$, and let N be a stopping time. Conditional on $\{N < \infty\}$, the sequence $(X_{N+n})_{n\geq 1}$ is independent of $\mathcal{F}_N$ and has the same distribution as the original sequence.

Example (Returns to 0)

Let $S = \mathbb{R}$ and

$$T(\omega) := \inf\{n \in \mathbb{N} : \omega_1 + \cdots + \omega_n = 0\}.$$

Define recursively

$$T_{k+1}(\omega) := T_k(\omega) + T(\theta^{T_k}\omega), \qquad T_0 = 0,$$

where θ is the shift operator.

Lemma 1

On the product space with $\mathbb{P} = \mu \times \mu \times \cdots$,

$$\mathbb{P}(T_n < \infty) = (\mathbb{P}(T < \infty))^n.$$

Lemma 2

Suppose $\mathbb{P}(T < \infty) = 1$ and let

$$t_n := T(\theta^{T_{n-1}}\omega).$$

Then

$$V_n = (t_n, X_{T_{n-1}+1}, \ldots, X_{T_n})$$

are i.i.d.

Example (Ladder variables)

Let

$$d(\omega) := \inf\{n \in \mathbb{N} : \omega_1 + \cdots + \omega_n > 0\}.$$

Let $d_0 = 0$ and

$$d_k = d_{k-1} + d(\theta^{d_{k-1}}\omega).$$

At time d_k the random walk is at the k–th record high value.

Wald's equations

1. Let (X_n) be i.i.d. with $\mathbb{E}|X_1| < \infty$. If N is a stopping time with $\mathbb{E}N < \infty$, then

$$\mathbb{E}(S_N) = \mathbb{E}(X_1)\mathbb{E}(N).$$

2. Let (X_n) be i.i.d. with $\mathbb{E}X_1^2 = \sigma^2 < \infty$ and $\mathbb{E}X_1 = 0$. If N is a stopping time with $\mathbb{E}N < \infty$, then

$$\mathbb{E}(S_N^2) = \sigma^2\,\mathbb{E}(N).$$

These will be obtained as applications of the optional stopping theorem for martingales. A direct proof can be given by splitting according to $\{N = n\}$ and using Fubini and independence, then extending by induction. See Durrett, Theorem 2.6.2.

References